

Sapna Tiwari

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github.com/sapna1475 | linkedin.com/in/sapna-tiwari | leetcode.com/u/sapna1475tiwari

Education

Vellore Institute of Technology (VIT)

2023 – 2027

B.Tech, Computer Science & Engineering

GPA: 9.03/10

Skills

Languages: Python, Java, JavaScript, React.js, HTML, CSS

Frameworks/Technologies: FastAPI, Streamlit, Node.js, ExpressJS, RESTful APIs, JWT, PostgreSQL, MongoDB, SQL

AI/LLM Engineering: LangChain, LangGraph, Multi-Agent Systems, Agentic AI, RAG Pipelines, Vector Databases (FAISS), Tool Calling, Structured Outputs, LLM Memory (Short-Term, Long-Term), Prompt Engineering, Hugging Face

LLM Evaluation & Observability: LangSmith, RAGAS, Evaluation Datasets, Tracing

Cloud & DevOps: AWS (ECR, ECS Fargate, Bedrock), Docker, Git/GitHub, CI/CD Pipelines (GitHub Actions), Render

Projects

Blog Writing Agent

github.com/sapna1475/Blog-Writing-Agent | *Live Demo (AWS)*

LangGraph, LangSmith, FastAPI, Streamlit, Docker, PostgreSQL, Hugging Face, Gemini

- Cut blog-generation latency 68% (40 to 13 min, 50 topics) by building a LangGraph multi-agent pipeline with parallel section-writer agents.
- Reduced citation hallucinations 92% and raised parse success 61% to 98% via Pydantic validation, and human-in-the-loop approval with Postgres-backed short-term memory.
- Improved citation-validity rate from 74% to 93% by tracing every run with LangSmith; containerized with Docker and deployed on AWS (Amazon ECR, ECS Fargate).

YouTube RAG Assistant

github.com/sapna1475/RAG-YOUTUBE-CHAT | youtube-rag-backend-mvqq.onrender.com/docs

FastAPI, LangChain, FAISS, HuggingFace, RAGAS, JavaScript, HTML

- Delivered transcript-grounded Q&A across 500+ YouTube videos by combining FAISS vector search, embeddings, and Qwen 2.5 7B inference in a full-stack RAG system.
- Achieved 70% Precision@2, 81.1% faithfulness, and 78.5% answer relevancy (RAGAS) by building an evaluation pipeline on 100 hand-labeled questions.
- Cut repeated-query latency 85% (20s to 3s) by caching per-video FAISS indexes across 24,832 transcript chunks.

AI Habit Tracker

github.com/sapna1475/AI-HabitTracker

Google Gemini 2.5 Flash, MERN Stack, PWA, VAPID, JWT, bcrypt

- Delivered sub-2-second responses across 5 Gemini 2.5 Flash features by applying Express rate-limiting middleware.
 - Sustained 94.6% success across 300+ simulated AI requests by adding circuit-breaker logic and automatic fallback recovery.
 - Eliminated duplicate notifications (0 across 1,000 simulated users) by building an idempotent node-cron and VAPID Web Push system with per-user-per-day state checks.
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Activities & Leadership

- **VIT Bhopal Cyber Security Club** – Event Management Lead, Cyber Carnival '26 (Dec 2025 – Jan 2026): drove 1,000+ student registrations via social media campaigns and college partnerships.
- **ICDCC 2024** – Web Development Team Member (Nov – Dec 2024): built the homepage and speaker section of a conference site serving 5,000+ visits.
- **Certifications:** Google IT Support Certificate (credly.com/go/snA6axFV); Deloitte Data Analytics Job Simulation.
- **Interests:** Chess, Swimming, Sketching, Badminton.